



USA TRIATHLON EDUCATION

7-Zone Training System for Cycling

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Overview

The 7-zone system anchors every cycling zone to LT1 and LT2, which on the bike is FTP. Zones 1 and 2 sit at or below LT1, Zones 3 and 4 sit between LT1 and LT2, and Zones 5 to 7 sit above FTP.

Cycling zones are usually written as a percentage of FTP, so they map straight into power-based tools. The athlete's LT1 percentage still moves the lower zones, and that is what separates the three athlete categories below.

The seven cycling zones

Zone	Power definition	Position	Purpose
1	80% to 90% of LT1 power	Below LT1	Recovery and easy spinning
2	90% to 100% of LT1 power	Up to LT1	Aerobic base: capillarization, blood volume, fat use
3	LT1 power to the midpoint of LT1 and LT2	Between LT1 and LT2	Tempo, moderate aerobic work
4	Midpoint of LT1 and LT2 up to LT2 (FTP)	Just below LT2	Sub-threshold, sweet spot, threshold building
5	100% to 107% of LT2	Just above LT2	Above-threshold work
6	107% to 118% of LT2	Higher intensity	Anaerobic capacity, lactate tolerance, VO2 work
7	118% of LT2 up to the Zone 7 ceiling	Near maximal	Neuromuscular power, sprints

Percent of LT2 (FTP) power by athlete category:

Zone	Age-Group (LT1 = 70%)	Performance (LT1 = 75%)	Elite (LT1 = 80%)
1	56 - 63%	60 - 67.5%	64 - 72%
2	63 - 70%	67.5 - 75%	72 - 80%
3	70 - 85%	75 - 87.5%	80 - 90%
4	85 - 100%	87.5 - 100%	90 - 100%
5	100 - 107%	same	same
6	107 - 118%	same	same
7	118 - 150%*	same	same

*Zone 7 tops out at the Zone 7 ceiling, which defaults to 150% of LT2. In manual mode, it uses the peak power entered instead.

Zones 5 to 7 are fixed multiples of LT2, so they are the same for every athlete. Zones 1 to 4 shift with LT1, which is about 70% of LT2 for Age-Group, 75% for Performance, and 80% for Elite. Zones 3 and 4 split the gap between LT1 and LT2 at the midpoint. If you use a full lactate test instead of an estimate, use the athlete's actual LT1 divided by LT2 in place of the category percentage.

For heart rate, Zones 1 to 4 follow the same structure, with LT1 heart rate at 83, 86, or 89% of threshold heart rate depending on category. Zones 5 to 7 are built from the gap between threshold heart rate and maximum heart rate. Remember that heart rate lags power and moves with heat, hydration, and stress.

Cadence and the zones

The same power at a different cadence is a different stimulus. At 60 rpm the muscular system dominates and heart rate response is lower. At 110 rpm the neuromuscular demand and heart rate are higher. So 200 W at 60 rpm and 200 W at 110 rpm are not the same workout.

- **Know the athlete's tendency.** Athletes with strong muscular systems drift to low cadence, and those with better cardiovascular systems pick high cadence.
- **Prescribe against the weakness.** Someone sitting near 70 rpm benefits from cadence development, and someone at 110 rpm needs strength work.
- **Periodize it.** Low-cadence strength work (50s to 70s rpm) fits base phases and pairs well with Zone 2 aerobic development. Race-specific cadence work belongs closer to competition.

Work-to-rest ratios for cycling

The harder the interval, the longer the rest it needs. Insufficient rest also degrades pedaling mechanics and the quality of the next effort.

Zones	Energy system	Work-to-rest guidance for cycling
1-2	Aerobic	Minimal byproducts and quick recovery. Long steady rides, with 40 to 70 mile rides for long-course base
3-4	Aerobic into anaerobic	Rest lengthens as power approaches FTP. Threshold-and-hold intervals can extend from 10 minutes to 16 minutes or more to build fatigue resistance. Sweet spot (88 to 93% of FTP) improves lactate clearance below threshold
5-6	Anaerobic	Substantially more rest so lactate can partially clear. Example race-specific set: 1 minute at FTP then 8 minutes at race pace, repeated 4 times with 3 to 4 minutes of recovery. Maximal efforts only (make sure athletes have been signed off by their doctor for physical exercise, and only complete once the athlete is cleared and conditioned once the athlete is well conditioned to training)
7	Alactic / ATP-CP	Sprints under 10 seconds need near-full recovery, around 1:5 to 1:8 work to rest. Short-course athletes can add 15-second sprints after threshold work

Anaerobic capacity sessions are worth including, but about one every two weeks is enough to maintain the system without overtraining.

Coaching notes for cycling

- **Match the zones to race demands.** Draft-legal athletes need Zones 5 to 7 for repeated surges. Long-course athletes spend most of their time in Zones 2 to 4 and need fatigue resistance.
- **Look past FTP.** An athlete can get faster without changing FTP for months by improving fatigue resistance. Track functional reserve capacity and fatigue rates as well.
- **Retest on schedule.** Re-establish FTP every 3 to 6 months and reset zones after each test, in the first week after an adaptation week.
- **Use the right test.** A 20-minute test suits beginner or remote athletes. A step test suits performance athletes and gives richer data.